Dr Andrew Mummery

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School of Natural Sciences
Institute for Advanced Study, Princeton, NJ
GitHub *Website*

Employment

2025–present **John N. Bahcall Fellow and Member**, *School of Natural Sciences*, Institute for Advanced Study.
Independent research on black hole accretion, tidal disruption events, and strong-gravity astrophysics.

2022–2025 **Leverhulme-Peierls Fellow**, *Rudolf Peierls Centre for Theoretical Physics*, University of Oxford.
Independent research fellowship in black hole accretion physics.

2022–2025 **Junior Research Fellow**, *Merton College*, University of Oxford.
Non-stipendiary junior research fellowship.

Education

2018–2022 **DPhil, Theoretical Physics**, *University of Oxford*, Wadham College.
Thesis: *Illuminating tidal disruption events with a time-dependent theory of relativistic accretion discs*
Supervisor: Prof. Steven Balbus
Thesis awarded best-thesis prizes by the International Astronomical Union and the Royal Astronomical Society, and an "exceptional" letter of commendation by the University of Oxford.

2014–2018 **Master of Physics**, *Department of Physics*, University of Oxford, Wadham College.
First-class honours; highest degree mark in the year group. The final-year Master's project was subsequently published in the Monthly Notices of the Royal Astronomical Society.

Fellowships & Awards

2025–2030 **John N. Bahcall Fellowship**, *Institute for Advanced Study*.

2022–2025 **Leverhulme-Peierls Fellowship**, *University of Oxford*.

2022 **International Astronomical Union best thesis prize**, *Division D: Fundamental Physics and High Energy Astrophysical Phenomena*.

2022 **Royal Astronomical Society Michael Penston thesis prize**.

2022 **Non-stipendiary junior research fellowship**, Merton College, Oxford University.

2022 **Letter of commendation for an "exceptional" DPhil thesis**, Oxford University.

2020 **Visiting research scholarship**, Princeton University.

2019 **Keeley graduate scholarship**, Wadham College, Oxford University.

2018 **Scott prize**, Oxford University.
Awarded for the top Physics degree mark in the class of 2018

2018 **Sukumar prize**, Wadham College, Oxford University.
Awarded for the top Physics degree mark in the class of 2018

Students

PhD **Muryel Guolo** (Johns Hopkins → IAS). Papers led with **Mummery**:
Guolo, **Mummery**, van Velzen et al. (2026), The wandering supermassive black hole powering the off-nuclear TDE AT2024tvd. ApJL 1004, L25. arXiv.
Guolo, **Mummery**, van Velzen et al. (2025), Compact accretion disks in the aftermath of tidal disruption events: parameter inference from joint X-ray spectra and UV/optical photometry. Submitted to ApJ. arXiv.
Guolo, **Mummery**, Ingram, Nicholl, Gezari & Nathan (2025), A time-dependent solution for GSN 069 disk evolution and the nature of long-lived tidal disruption events. ApJ 992, 114. arXiv.
Guolo, **Mummery**, Wevers, Nicholl, Gezari, Ingram & Pasham (2025), The properties of the GSN 069 accretion disk from a joint X-ray and UV spectral analysis: stress-testing quasi-periodic eruption models. ApJ 985, 146. arXiv.
Guolo & **Mummery** (2025), The size of accretion disks from self-consistent X-ray spectra and UV/optical/NIR photometry fitting: ASASSN-14li and HLX-1 — diskSED. ApJ 978, 167. arXiv.

PhD **Jake Rule** (University of Oxford; on the job market). Selected papers:
Rule, **Mummery**, Balbus, Stone & Zhang (2025), The plunging region of a thin accretion disc around a Schwarzschild black hole. MNRAS 542, 377. arXiv.
Rule, **Mummery**, Balbus, Stone & Zhang (2026), The plunging region of thin accretion discs across the black hole spin range. Submitted to MNRAS. arXiv.

PhD **Fangyuan (Jason) Yu** (Princeton University). Papers led with **Mummery**:
Yu, **Mummery** & Guolo (2026), HLX–1 as a repeating partial tidal disruption event around an intermediate-mass black hole. arXiv.

Summer **Wenkang Xin**. Xin & **Mummery** (2026), The relativistic tidal tensor: general solutions for stationary axisymmetric spacetimes and the Hills mass of naked singularities. Physical Review D 113, 044036. arXiv. Now a student with John Magorrian at the University of Oxford.

Observing programmes

2025– **Hubble Space Telescope GO 18081 (Principal Investigator)**.
Using 35-year-old tidal disruption events to test the fundamentals of accretion disk theory. Award: $60,000.

Cycle 5 **JWST GO 11908 (co-PI)**.
Probing the M–σ Relation in the Intermediate-Mass Black Hole Regime with Off-Nuclear TDEs. Award: $175,000.

Cycle 5 **JWST GO 12397 (co-PI)**.
Where Are the Stars? A Search for the Stellar Counterparts of Two Off-Nuclear TDE Candidates. Award: $110,000.

Cycle 8 **NICER C8 9062 (co-PI)**.
A Time-Dependent View of XRB Outbursts: Spectroscopy Beyond Steady-State Models. Award: $60,000.

Teaching and Supervising

2024–2025 **Lecturer: Magnetohydrodynamics**, University of Oxford.
I wrote and gave 10 one-hour lectures covering the fundamentals of MHD through to topics of modern research. This course is attended by Masters and PhD students in the Mathematics and Physics departments. I then wrote, set and marked an exam paper for the Masters students. Lecture notes available here.

2018-2022 **Problem classes – (College Tutor)**, University of Oxford (various colleges),
– General Relativity (3rd year undergraduate)
– Special Relativity (3rd year undergraduate)
– Chaos, Random Processes and Predictability (2nd year undergraduate)
– Fluid Dynamics (3rd year undergraduate).
Led small group problem classes (typically 2-5 students), marked and reviewed mock exam papers, and led final exam revision sessions for larger classes (typically 8-10 students).

2021–2024 **Undergraduate interviews**, University College, University of Oxford.
Interviewed roughly 50 prospective undergraduates each year; wrote physics and mathematics questions and served on the College shortlisting and selection panel.

Peer review

2019–present **Journal refereeing**.
Peer reviewer for the Monthly Notices of the Royal Astronomical Society, The Astrophysical Journal, The Astrophysical Journal Letters, The Open Journal of Astrophysics, Physical Review D, Astronomy & Astrophysics, and Astroparticle Physics.

2024 **PhD examining**.
External PhD examiner for the thesis defence of Jesse Voss at the University of Radboud.

Outreach and media

2026 **Media engagement**.
Interviewed by CTV News, with related coverage in *The Guardian*, on black hole accretion and jet formation.

2024 **Media engagement**.
For our paper titled "Continuum emission from within the plunging region of black hole discs" I undertook a round of media interviews (including two Youtube interviews gaining over half a million views), to amplify the impact of our work. This paper ended up being the 60th most-discussed paper in the journals history. In total over 200 pieces of coverage reaching a global audience in excess of 150 million people were published, including the following:
Scientific American, New Scientist, The Times, CNN, The Daily Mail, The Times of India, NASA Today, MSN News, The Independent, The Evening Standard, Mirage News, Associated Press, BBC and Newsweek.

2023–2024 **Public lectures**.
– *"The spaghettification of stars by supermassive black holes: understanding one of nature's most extreme events"* Public lecture for the event series "A Saturday Morning of Theoretical Physics" (podcast available).
– *"Taking the plunge: detecting light from the very edge of black holes"* Public lecture to the Guildford Astronomy Society.

2018–2022 **Outreach – UNIQ summer school tutor**.
I ran problem classes for underprivileged high school students from around the UK who visit Oxford for a week, with the aim of encouraging them to apply to Universities in the future

Software

FitTeD GitHub organisation for fitting transients with relativistic discs. Personal repositories are being migrated here.

FitTeD package Python package for light curves and spectra of evolving relativistic accretion discs, including **TDEFLARE** for rapid black hole masses from TDE optical/UV flares. code, paper.

tidalspin Bayesian black hole spin constraints from the Hills-mass limit in TDEs. code, paper.

XSPEC models Public spectral models: fullkerr, TDEdiscXraySpectrum, superkerr, superkerrline, GR_prop_fluc.

Publications

76 publications; 42 first-author publications. Papers with A. Goodwin or J. Chakraborty marked * are joint first-author. † denotes a paper led by a student. Full, current lists: NASA ADS, Google Scholar, and arXiv. Listed in reverse chronological order.

[76]. Chakraborty*, **Mummery***, Quataert, Arcodia, Linial & Kara (2026), Quasi-periodic eruption spectral-timing: EMRIs crossing warped accretion disks. Submitted. arXiv. *Joint first authors.

[75]. †Yu, **Mummery** & Guolo (2026), HLX–1 as a repeating partial tidal disruption event around an intermediate-mass black hole. arXiv

[74]. **Mummery** & Quataert (2026), Observational signatures of warped accretion flows in tidal disruption events. Submitted to ApJ. arXiv

[73]. **Mummery** & Quataert (2026), The chirping of Lense–Thirring precession in tidal disruption event accretion flows. Submitted to ApJ. arXiv

[72]. †Rule, **Mummery**, Balbus, Stone & Zhang (2026), The plunging region of thin accretion discs across the black hole spin range. Submitted to MNRAS. arXiv

[71]. Goodwin, Miller-Jones, Sfaradi, **Mummery** et al. (2026), One year of broadband radio monitoring of the enigmatic transient GRB 250702B reveals the evolution of the relativistic jet. Submitted to MNRAS. arXiv

[70]. **Mummery**, Goodwin, Christy, Alexander & Franz (2026), Triple radio flares from tidal disruption events: jet–wind collisions and the discovery of a third radio flare from AT2020vwl. Submitted. arXiv

[69]. Li, Christy, Alexander, Sfaradi, Huang, Jiang, Laskar, **Mummery** et al. (2026), VLA observations confirm AT 2023mfm as an off-nuclear tidal disruption event. RNAAS 10, 153. arXiv

[68]. Arcodia, Miniutti, Chakraborty, Franchini, Giustini, Linial, **Mummery** et al. (2026), Even a precessing clock is right twice per orbit — the super-periods of eRO-QPE2 and challenges for QPE orbital models. ApJ 1003, 148. arXiv

[67]. Goodwin* & **Mummery*** (2026), Resolving the sub-parsec circumnuclear density profiles of quiescent galaxies: evidence for Bondi accretion flows in TDE hosts. Submitted to ApJ. arXiv. *Joint first authors.

[66]. Yao, Chornock, **Mummery** et al. (2026), AT2024lhc and AT2024kmq in the landscape of featureless tidal disruption events. MNRAS 549, stag920. arXiv

[65]. **Mummery** (2026), Accretion disks in (repeating) partial tidal disruption events: rapid state transitions, UV plateaus and flares from disk–remnant collisions. MNRAS 547, stag387. arXiv

[64]. Chakraborty, Masterson, **Mummery**, Kara, Panagiotou, Arcodia & Berger (2026), X-ray spectral-timing properties of tidal disruption events. ApJ 1000, 95. arXiv

[63]. Goodwin* & **Mummery*** (2026), A universal critical accretion rate for black hole jet formation. Nature Astronomy. arXiv. *Joint first authors.

[62]. †Guolo, **Mummery**, van Velzen et al. (2026), The wandering supermassive black hole powering the off-nuclear TDE AT2024tvd. ApJL 1004, L25. arXiv

[61]. Hamilton, **Mummery** & Bland-Hawthorn (2026), On the radial velocity wave in the Galactic disk. Submitted to ApJ. arXiv

[60]. **Mummery** (2026), Supermassive black hole mass inference with the optical flares of tidal disruption events — TDEFLARE. Submitted. arXiv

[59]. **Mummery** (2026), Black hole accretion disks with outflows II. Time-dependent Green's function solutions in Newtonian gravity. Submitted. arXiv

[58]. Omand, Sarin, Lamb, Perley, **Mummery** et al. (2026), Multiwavelength modeling of the luminous fast blue optical transient AT2024wpp. MNRAS 550, stag1104. arXiv

[57]. †Xin & **Mummery** (2026), The relativistic tidal tensor: general solutions for stationary axisymmetric spacetimes and the Hills mass of naked singularities. Physical Review D 113, 044036. arXiv

[56]. Chrimes, Jonker, Levan & **Mummery** (2026), Luminous fast blue optical transients as very massive star core-collapse events. A&A 706, A327. arXiv

[55]. **Mummery**, Metzger, van Velzen & Guolo (2025), Tidal disruption event calorimetry: observational constraints on the physics of TDE optical flares. Submitted. arXiv

[54]. †Guolo, **Mummery**, van Velzen et al. (2025), Compact accretion disks in the aftermath of tidal disruption events: parameter inference from joint X-ray spectra and UV/optical photometry. Submitted to ApJ. arXiv

[53]. Inkenhaag, Levan, **Mummery** & Jonker (2025), AT 2018cow at ∼5 years: additional evidence for a tidal disruption origin. MNRAS 544, L108. arXiv

[52]. Saxton, Wevers, van Velzen, Alexander, Liu, **Mummery** et al. (2025), Rapid onset of a Comptonisation zone in the repeating tidal disruption event XMMSL2 J140446.9-251135. A&A 704, A165. arXiv

[51]. Ramsden, Nicholl, McGee & **Mummery** (2025), Evidence for a steeper SMBH–bulge mass relationship extended to low masses using TDE host galaxies. MNRAS 541, 1218. arXiv

[50]. **Mummery** & Silk (2025), Black hole supercolliders. Physical Review Letters 134, 221401. arXiv

[49]. †Rule, **Mummery**, Balbus, Stone & Zhang (2025), The plunging region of a thin accretion disc around a Schwarzschild black hole. MNRAS 542, 377. arXiv

[48]. **Mummery**, Jiang, Ingram, Fabian & Rule (2025), A rapid black hole spin or emission from the plunging region? MNRAS 544, 2880. arXiv

[47]. **Mummery** (2025), The X-ray variability of tidal disruption events. MNRAS 544, 1627. arXiv

[46]. **Mummery** (2025), Collisions with tidal disruption event disks: implications for quasi-periodic X-ray eruptions. Under review, MNRAS. arXiv

[45]. †Guolo, **Mummery**, Ingram, Nicholl, Gezari & Nathan (2025), A time-dependent solution for GSN 069 disk evolution and the nature of long-lived tidal disruption events. ApJ 992, 114. arXiv

[44]. Chakraborty, Kara, Arcodia, Buchner, Giustini, Hernández-García, Linial, Masterson, Miniutti, **Mummery** et al. (2025), Discovery of quasi-periodic eruptions in the TDE and extreme coronal line emitter AT2022upj. ApJL 983, L39. arXiv

[43]. **Mummery**, Guolo, Matthews, Newsome, Lintott & Keel (2025), Galaxy-scale consequences of tidal disruption events: extended emission line regions, extreme coronal lines and infrared-to-optical light echoes. MNRAS 544, 2262. arXiv

[42]. **Mummery** (2025), Black hole–disc coevolution in the presence of magnetic fields: refining the Thorne limit with emission from within the plunging region. MNRAS 537, 1963. arXiv

[41]. Wevers, Guolo, Lockwood, **Mummery**, Pasham & Arcodia (2025), Time-resolved Hubble Space Telescope UV observations of an X-ray quasi-periodic eruption source. ApJL 980, L1. arXiv

[40]. †Guolo, **Mummery**, Wevers, Nicholl, Gezari, Ingram & Pasham (2025), The properties of the GSN 069 accretion disk from a joint X-ray and UV spectral analysis: stress-testing quasi-periodic eruption models. ApJ 985, 146. arXiv

[39]. Witte & **Mummery** (2025), Stepping up superradiance constraints on axions. Physical Review D 111, 083044. arXiv

[38]. Goodwin, **Mummery**, Laskar, Alexander et al. (2025), A second radio flare from the tidal disruption event AT2020vwl: a delayed outflow ejection? ApJ 981, 122. arXiv

[37]. **Mummery** & van Velzen (2025), The optical, UV-plateau and X-ray tidal disruption event luminosity functions reproduced from first principles. MNRAS 541, 429. arXiv

[36]. Bright, Carotenuto, Fender, Choza, **Mummery** et al. (2025), The radio counterpart to the fast X-ray transient EP240414a. ApJ 981, 48. arXiv

[35]. †Guolo & **Mummery** (2025), The size of accretion disks from self-consistent X-ray spectra and UV/optical/NIR photometry fitting: ASASSN-14li and HLX-1 — diskSED. ApJ 978, 167. arXiv

[34]. **Mummery**, Nathan, Ingram & Gardner (2025), Fitting transients with discs — FitTeD. MNRAS 544, 2225. arXiv

[33]. Nicholl, Pasham, **Mummery** et al. (2024), Quasi-periodic X-ray eruptions years after a nearby tidal disruption event. Nature 634, 804. arXiv

[32]. **Mummery** & Stone (2024), The three-dimensional structure of black hole accretion flows within the plunging region. MNRAS 532, 3395. arXiv

[31]. **Mummery**, Jiang & Fabian (2024), Plunging region emission in the X-ray binary MAXI J0637-430. MNRAS 533, L83. arXiv

[30]. Chrimes, Coppejans, Jonker, Levan, Groot, **Mummery** & Stanway (2024), Multi-wavelength observations of the luminous fast blue optical transient AT2023fhn. A&A 691, A329. arXiv

[29]. **Mummery**, Ingram, Davis & Fabian (2024), Continuum emission from within the plunging region of black hole discs. MNRAS 531, 366. arXiv

[28]. **Mummery** & Turner (2024), The turbulent variability of accretion discs observed at high energies. MNRAS 530, 4730. arXiv

[27]. Pasham, Tombesi, Suková, Zajaček, Rakshit, Coughlin, Kosec, Karas, Masterson, **Mummery** et al. (2024), A case for a binary black hole system revealed via quasi-periodic outflows. Science Advances 10, eadj8898. arXiv

[26]. **Mummery**, Mori & Balbus (2024), The dynamics of accretion flows near to the innermost stable circular orbit. MNRAS 529, 1900. arXiv

[25]. **Mummery** & Ingram (2024), Reflecting on naked singularities: iron line fitting as a probe of the cosmic censorship conjecture. MNRAS 528, 2015. arXiv

[24]. **Mummery** (2024), The maximum mass of a black hole which can tidally disrupt a star: measuring black hole spins with tidal disruption events — tidalspin. MNRAS 527, 6233. arXiv

[23]. **Mummery**, Balbus & Ingram (2024), Testing theories of accretion and gravity with super-extremal Kerr discs. MNRAS 527, 5956. arXiv

[22]. **Mummery**, van Velzen, Nathan, Ingram, Hammerstein, Fraser-Taliente & Balbus (2024), Fundamental scaling relationships revealed in the optical light curves of tidal disruption events. MNRAS 527, 2452. arXiv

[21]. Chrimes, Jonker, Levan, Coppejans, Gaspari, Gompertz, Groot, Malesani, **Mummery** et al. (2024), AT2023fhn (the Finch): a luminous fast blue optical transient at a large offset from its host galaxy. MNRAS 527, L47. arXiv

[20]. Inkenhaag, Jonker, Levan, Chrimes, **Mummery**, Perley & Tanvir (2023), Late-time HST UV and optical observations of AT 2018cow: extracting a cow from its background. MNRAS 525, 4042. arXiv

[19]. **Mummery** (2023), Extending the theory of propagating fluctuations: the first fully relativistic treatment and analytical Fourier–Green's functions. MNRAS 523, 3629. arXiv

[18]. Wiseman, Wang, Hönig, Castro-Segura, Clark, Frohmaier, Fulton, Leloudas, Middleton, Müller-Bravo, **Mummery** et al. (2023), Multiwavelength observations of the extraordinary accretion event AT2021lwx. MNRAS 522, 3992. arXiv

[17]. **Mummery** & Balbus (2023), Accretion within the innermost stable circular orbit: analytical thermodynamic solutions in the adiabatic limit. MNRAS 521, 2439. arXiv

[16]. **Mummery** & Balbus (2023), A complete characterisation of the orbital shapes of the non-circular Kerr geodesic solutions with circular orbit constants of motion. Physical Review D 107, 124058. arXiv

[15]. **Mummery**, Wevers, Saxton & Pasham (2023), From X-rays to physical parameters: a comprehensive analysis of thermal tidal disruption event X-ray spectra. MNRAS 519, 5828. arXiv

[14]. **Mummery** (2023), Asymptotic Green's function solutions of the general relativistic thin disc equations. MNRAS 518, 1905. arXiv

[13]. **Mummery** & Balbus (2022), The high energy probability distribution of accretion disc luminosity fluctuations. MNRAS 517, 3423. arXiv

[12]. **Mummery** & Balbus (2022), Inspirals from the innermost stable circular orbit of Kerr black holes: exact solutions and universal radial flow. Physical Review Letters 129, 161101. arXiv

[11]. Goodwin, van Velzen, Miller-Jones, **Mummery** et al. (2022), AT2019azh: an unusually long-lived, radio-bright thermal tidal disruption event. MNRAS 511, 5328. arXiv

[10]. **Mummery** (2021), Tidal disruption event discs are larger than they seem: removing systematic biases in TDE X-ray spectral modelling. MNRAS 507, L24. arXiv

[9]. **Mummery** (2021), A unified model of tidal destruction events in the disc-dominated phase. arXiv. arXiv

[8]. **Mummery** (2021), A maximum X-ray luminosity scale of disc-dominated tidal disruption events. MNRAS 504, 5144. arXiv

[7]. **Mummery** & Balbus (2021), Hard X-ray emission from a Compton scattering corona in large black hole mass tidal disruption events. MNRAS 504, 4730. arXiv

[6]. **Mummery** & Balbus (2021), An upper observable black hole mass scale for tidal disruption events with thermal X-ray spectra. MNRAS 505, 1629. arXiv

[5]. **Mummery** & Balbus (2020), ASASSN-15lh: a TDE about a maximally rotating $10^9 M_\odot$ black hole. MNRAS 497, L13. arXiv

[4]. **Mummery** & Balbus (2020), The spectral evolution of disc-dominated tidal disruption events. MNRAS 492, 5655. arXiv

[3]. **Mummery** & Balbus (2019), Evolution of relativistic thin discs with a finite ISCO stress: II. Late time behaviour. MNRAS 489, 143. arXiv

[2]. **Mummery** & Balbus (2019), Evolution of relativistic thin discs with a finite ISCO stress: I. Stalled accretion. MNRAS 489, 132. arXiv

[1]. Balbus & **Mummery** (2018), The evolution of Kerr discs and late-time tidal disruption event light curves. MNRAS 481, 3348. arXiv

Selected invited talks (2026)

October	Astronomy colloquium, University of California, Berkeley, USA.
August	Chandra Dynamic Universe meeting, Boston, USA.
August	COSPAR, Florence, Italy.
August	TDE conference, Crete, Greece.
July	Dutch Black Hole Consortium meeting, Amsterdam, the Netherlands.
June	X-ray Universe meeting, Elche, Spain.
April	Perimeter Institute, Waterloo, Canada.
April	New York University, New York, USA.
February	TDE workshop, Kyoto, Japan.